and the initial condition that $y = 10$ when $x = 2$.
From the given equation

$$\frac{dy}{dx} = 2x$$
$$y = \int 2x\,dx$$
$$y = x^2 + c$$

We substitute $y = 10$ when $x = 2$, in the above equation

$$10 = 2^2 + c \;\Rightarrow\; c = 6$$

When this value c =6 is substituted we obtain

$$y = x^2 + 6$$

which gives the particular antiderivative desired.

11.3 Basic Rules of Integration

Standard results:

Since integration is the reverse process of differentiation, the basic integration formulae given below can be derived directly from their corresponding derivative formulae from earlier chapter.

Derivatives	Antiderivatives
$\frac{d}{dx}(c) = 0,$ where c is a constant	$\int 0\,dx = c$, where c is a constant
$\frac{d}{dx}(kx) = k,$ where k is a constant	$\int k\,dx = kx + c$ where c is a constant
$\frac{d}{dx}\left(\frac{x^{n+1}}{n+1}\right) = x^n$	$\int x^n dx = \frac{x^{n+1}}{n+1} + c, \quad n \neq -1$ (Power rule)
$\frac{d}{dx}(\log x) = \left(\frac{1}{x}\right)$	$\int \frac{1}{x}\,dx = \log\lvert x\rvert + c$
$\frac{d}{dx}(-\cos x) = \sin x$	$\int \sin x\,dx = -\cos x + c$
$\frac{d}{dx}(\sin x) = \cos x$	$\int \cos x\,dx = \sin x + c$
$\frac{d}{dx}(\tan x) = \sec^2 x$	$\int \sec^2 x\,dx = \tan x + c$

11_11th-Maths-Vol2-Unit11.indd 186 23-03-2019 10:14:23